REPORT

OF

MACHINE LEARNING, DEEP LEARNING AND GENERATIVE AI WITH RETRIEVAL-AUGMENTED GENERATION (RAG)

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Roll Number: CS-23411167
Name of Student: GAURAV KUMAR SINGH
Batch: 2023-27

2026-27

CANDIDATE'S DECLARATION

I, GAURAV KUMAR SINGH, do hereby declare that the internship report titled "MACHINE LEARNING, DEEP LEARNING AND GENERATIVE AI WITH RETRIEVAL-AUGMENTED GENERATION (RAG)" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: GAURAV KUMAR SINGH

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my one-month internship in Artificial Intelligence and Generative AI at Ybi Foundation. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during the course of this work. I am sincerely grateful to the mentors and the training team at Ybi Foundation for sharing their truthful and illuminating views on Machine Learning, Deep Learning, Generative AI and Retrieval-Augmented Generation (RAG), which formed the core of this internship.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me the opportunity to undertake this internship as part of my B.Tech CSE curriculum, and for the continuous encouragement and support extended to me.

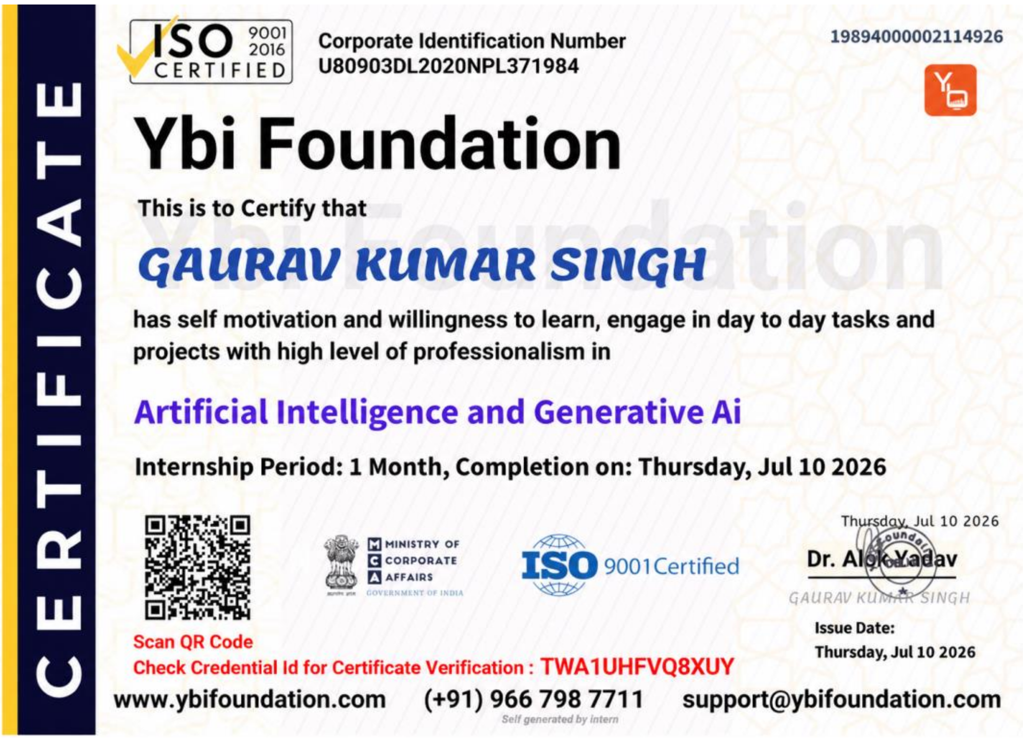
Finally, I express my heartfelt gratitude to my parents and friends for their constant blessings, patience and encouragement throughout this internship.

Signature

Student Name: GAURAV KUMAR SINGH

Roll No.: CS-23411167

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE

ISO 9001 2016 CERTIFIED

Corporate Identification Number
U80903DL2020NPL371984

19894000002114926

Ybi Foundation

This is to Certify that

GAURAV KUMAR SINGH

has self motivation and willingness to learn, engage in day to day tasks and projects with high level of professionalism in

Artificial Intelligence and Generative Ai

Internship Period: 1 Month, Completion on: Thursday, Jul 10 2026

Ministry of Corporate Affairs
Government of India

ISO 9001Certified

Thursday, Jul 10 2026
Dr. Alok Yadav
GAURAV KUMAR SINGH

Scan QR Code
Check Credential Id for Certificate Verification : TWA1UHFVQ8XUY

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Self generated by intern

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4. Project Description

4.1 Introduction

This report presents the work carried out during a one-month internship undertaken in the domain of Artificial Intelligence and Generative AI at Ybi Foundation. The internship was designed to build a progressive understanding of intelligent systems, beginning with the statistical foundations of Machine Learning (ML), advancing into Deep Learning (DL) and neural network architectures, and culminating in modern Generative AI techniques, including Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).

Machine Learning refers to the branch of computer science in which algorithms are trained on data to identify patterns and make predictions or decisions without being explicitly programmed for every rule. During the internship, foundational ML concepts such as supervised learning, unsupervised learning, regression, classification, and model evaluation metrics (accuracy, precision, recall, F1-score) were studied and implemented using Python and libraries such as scikit-learn, pandas and NumPy.

Deep Learning, a specialised subfield of Machine Learning, uses multi-layered artificial neural networks to automatically learn hierarchical feature representations from raw data such as images, text and audio. The internship covered the architecture and working of Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs) for image-related tasks, and Recurrent Neural Networks (RNNs)/LSTMs for sequential data, along with concepts such as backpropagation, activation functions, loss functions and optimisers (SGD, Adam).

The final and most contemporary phase of the internship focused on Generative AI — models capable of generating new content such as text, images and code rather than merely classifying or predicting on existing data. This included the study of Transformer architecture, the mechanism of self-attention, and Large Language Models such as the GPT family. Building on this, the internship introduced Retrieval-Augmented Generation (RAG), a technique that combines the generative power of LLMs with an external, up-to-date knowledge base retrieved at query time, thereby reducing hallucination and grounding responses in verifiable source documents.

4.2 Organization Profile

Ybi Foundation is an ISO 9001:2016 certified organisation registered under the Ministry of Corporate Affairs, Government of India, that provides structured internship and training programs to students in emerging technology domains such as Artificial Intelligence, Generative AI, Data Science and allied fields. The organisation focuses on bridging the gap between academic learning and industry-relevant skills through mentor-guided, project-based internship programs, self-paced modules and hands-on assignments, enabling interns to gain practical exposure to real-world AI workflows.

4.3 Problem Statement

Conventional academic curricula often emphasise theoretical foundations of Machine Learning and Deep Learning but provide limited exposure to how these concepts converge in modern Generative AI systems, particularly Large Language Models and Retrieval-Augmented Generation pipelines used extensively in the industry today. There was, therefore, a need to build a structured, hands-on understanding of the complete pipeline — from classical ML models to deep neural networks to transformer-based generative systems — and to understand how RAG architectures overcome the knowledge-cutoff and hallucination limitations of standalone LLMs by grounding generation in retrieved, verifiable data.

4.4 Project Objectives

- To understand and implement core Machine Learning algorithms, including supervised and unsupervised learning techniques, and to evaluate model performance using standard metrics.
- To study the architecture and training process of Deep Learning models, including ANNs, CNNs and RNNs/LSTMs, and to understand how they differ from classical ML approaches.
- To gain conceptual and practical understanding of Transformer architecture, self-attention and Large Language Models that power modern Generative AI applications.
- To understand the design and working of Retrieval-Augmented Generation (RAG) systems, including embeddings, vector databases and semantic retrieval.
- To build a small proof-of-concept application demonstrating the end-to-end RAG pipeline: document ingestion, chunking, embedding, retrieval and grounded response generation.
- To document the learning outcomes, challenges faced and practical skills gained during the internship period.

4.5 Scope of the Project

The scope of the internship was limited to the conceptual study, guided implementation and small-scale experimentation with ML, DL and Generative AI/RAG techniques over a period of one month. It included working with publicly available datasets and pre-trained models, building simple end-to-end demonstrations, and did not extend to deploying large-scale production systems, training foundation models from scratch, or handling proprietary/enterprise data. The RAG component was scoped to a proof-of-concept level, sufficient to demonstrate the underlying architecture and workflow rather than a production-grade deployment.

4.6 Technologies and Tools Used

- Programming Language: Python
- ML Libraries: scikit-learn, pandas, NumPy, Matplotlib/Seaborn
- Deep Learning Frameworks: TensorFlow / Keras and PyTorch
- Generative AI / LLM Tools: Hugging Face Transformers, OpenAI/LLM APIs
- RAG Components: Text embedding models, vector databases (e.g., FAISS/Chroma), LangChain-style orchestration
- Development Environment: Jupyter Notebook / Google Colab, VS Code
- Version Control: Git and GitHub

4.7 System Architecture

The overall learning-and-build architecture followed during the internship can be described in three progressive layers:

- Layer 1 – Machine Learning Layer: Raw structured data is pre-processed and fed into classical ML algorithms (e.g., Logistic Regression, Decision Trees, KNN) for prediction/classification tasks, with model evaluation performed using train-test splits and cross-validation.
- Layer 2 – Deep Learning Layer: Unstructured data (images/text) is passed through neural network architectures (CNN/RNN/LSTM), where multiple hidden layers extract hierarchical features, and the model is optimised using backpropagation and gradient-descent-based optimisers.
- Layer 3 – Generative AI / RAG Layer: User queries are converted into vector embeddings; a vector database performs a similarity search over embedded document chunks to retrieve the most relevant context; this retrieved context, together with the original query, is passed to a Large Language Model, which generates a final, grounded natural-language response.

This layered architecture illustrates the natural progression from statistical learning to representation learning and finally to retrieval-grounded generative systems.

4.8 Methodology

The internship followed a structured, module-wise methodology spread across four weeks:

- Week 1 – Machine Learning Foundations: Study of supervised/unsupervised learning, data pre-processing, feature engineering, and implementation of regression and classification models with performance evaluation.
- Week 2 – Deep Learning: Study of perceptrons, multi-layer neural networks, activation and loss functions, and hands-on implementation of CNNs for image classification and RNN/LSTM for sequence data.
- Week 3 – Generative AI Foundations: Study of Transformer architecture, self-attention, tokenization, and how Large Language Models are pre-trained and fine-tuned; experimentation with prompt engineering on pre-trained LLMs.
- Week 4 – Retrieval-Augmented Generation (RAG): Study of embedding generation, vector similarity search, chunking strategies, and construction of a small end-to-end RAG pipeline that retrieves relevant document context and passes it to an LLM to generate accurate, grounded answers.

Throughout the program, each module combined conceptual study with hands-on coding exercises, mentor-guided sessions and self-assessment tasks, culminating in a final review of the concepts learned.

4.9 Expected Outcomes

- A clear conceptual understanding of the relationship between Machine Learning, Deep Learning and Generative AI.
- Practical ability to implement and evaluate basic ML and DL models.
- Understanding of how Large Language Models generate text and their inherent limitations, such as hallucination and static knowledge cut-offs.
- Working knowledge of the Retrieval-Augmented Generation architecture and its advantages in producing accurate, source-grounded responses.
- Enhanced problem-solving, self-learning and technical documentation skills relevant to real-world AI project work.

4.10 Certificates of Completion

The internship completion certificate issued by Ybi Foundation, confirming successful completion of the one-month internship in Artificial Intelligence and Generative AI, is attached in Section 3 (Internship Completion Certificate) of this report.

5. Bibliography / References

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